

Krishnaa Vadivel | Embedded Systems Engineer | Firmware | FPGA | Robotics

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Profile

Embedded systems engineer with experience spanning firmware development, FPGA logic, robotics, instrumentation, and field validation. Background includes low-level development for sensing and logging platforms, board bring-up and debugging, and companion software for analysis and visualization around embedded hardware.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed embedded functionality for geological well tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class hardware platforms.
- Implemented sensor-logging and control-side logic and firmware for field-deployed systems operating in instrumentation-heavy environments.
- Performed oscilloscope-based debugging, bench validation, and field testing to verify hardware-software integration and system behavior.
- Supported selective PCB updates in Altium and related board-level engineering tasks as system requirements evolved.
- Built companion host-side engineering tools using CUDA and OpenGL for waveform processing, visualization, and diagnostic workflows around the embedded stack.
- Contributed additional internal software using C#, ASP-style services, and Microsoft SQL-backed data workflows.

Yale University

Embedded Systems Teaching Assistant / PhD Researcher

2021–present

- Supported embedded systems laboratories and student projects involving firmware, hardware interfaces, debugging practices, and instrumentation workflows.
- Continued broader engineering work in scientific computing and simulation while maintaining strong embedded-systems fluency.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on excited-state materials physics.

Selected Projects

EEG-Driven Robotic Hand

- Built an embedded robotic-hand project in undergraduate ECE work, integrating sensing, actuation, and control; related work was published in Circuit Cellar.

Maze Mapping Robot with Wireless Control

- Worked on a robotics project combining navigation logic, remote-station control, and wireless coordination in a course setting.

Single-Cycle MIPS Processor

- Implemented a small Verilog CPU and self-checking test bench to demonstrate digital design, datapath organization, and reproducible verification.

Matrix-Multiply Accelerator

- Built a compact Verilog accelerator block with clear start/done control behavior and a simulation-backed test flow.

Host-Side Diagnostic and Visualization Tooling

- Developed CUDA/OpenGL-based utilities for analyzing and visualizing waveform data associated with embedded sensing and logging systems.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing doctoral work; also served as an embedded systems teaching assistant.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Included advanced engineering coursework and embedded-adjacent systems work.

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Included embedded systems, robotics, and hardware-software integration foundations.

Selected Coursework

Embedded: Advanced Embedded Systems; laboratory-oriented firmware and systems work

Signals and Systems: Mathematics of Signals and Systems

Hardware: VLSI Design; Analog VLSI Design; digital and mixed engineering foundations

Supporting Theory: Semiconductor Physics; controls/robotics-adjacent systems exposure in ECE training

Technical Skills

Languages: C, C++, Python, Bash, JavaScript

Hardware: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, sensor logging, embedded control

Interfaces: SPI, I2C, UART, hardware-software integration workflows

Validation: Oscilloscopes, bench instrumentation, field testing, board bring-up, Altium update support

Software: Linux, CUDA, OpenGL, C#, ASP-style services, Microsoft SQL